

INTRODUCTION

CREATED BY: DIONTE CAPLETON

- This dashboard analyzes loan approval patterns across demographic and financial variables using SQL, Python (Pandas), and Power BI.
- The dataset includes 614 loan applicants and 13 features, with missing data fully cleaned and processed.
- Core objective: uncover which applicant traits most influence loan approval rates and help inform faster loan processing decisions.
- Visualizations highlight trends in income, credit history, gender, and property area distribution.

RESEARCH QUESTIONS

- What factors predict and influence loan approval
- Are there income disparities in loan approval?
- How does the property area affect loan approval rates?
- What trends exist in gender-related loan outcomes?

DATA MODELING PROCESS

DATA SOURCE: Kaggle loan dataset, 614 records, 13 columns

TOOLS USED:



PROCESS:

- Cleaned and transformed data using Python (missing values)
- Loaded data into Oracle SQL for queries
- Modeled relationships between applicants, income, credit history, and loan status
- Analyzed patterns with Power BI visuals:
- bar charts, pie chart, DAX cards



LOAN DASHBOARD

MOST COMMON TERM

Loan_Amou...	Count of...
360	526
Total	526

GRADUATES APPLICANTS

480

MEDIAN LOAN AMOUNT

\$128

PROPERTY AREA

Property_Area	Count of Loan_ID
Semiurban	233
Total	233

KEY PROJECT METRICS

GENDER

Female

Male

TOTAL APPLICANTS

614

APPROVAL RATE

69%

MISSING DATA CLEANED

100% Missing Data Handled (7 columns)

DATA SIZE

614 Rows x 13 Columns

AVERAGE INCOME

4.81K

CREDIT HISTORY SPLIT

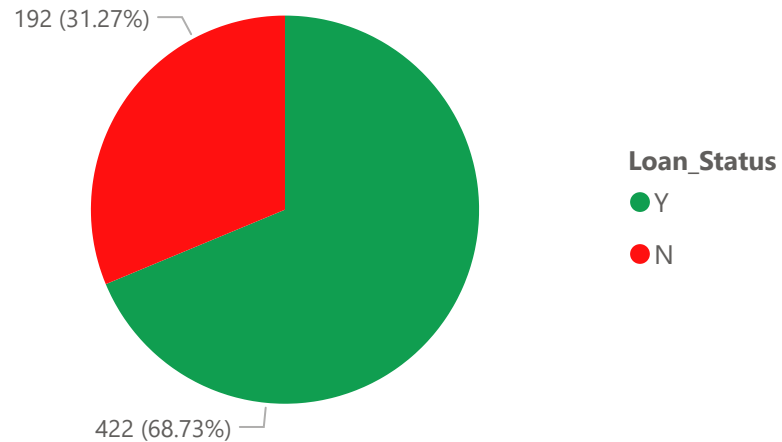
- Credit History = 1 → Good credit → High approval
- Credit History = 0 → Bad credit → Very low approval

LOAN INSIGHTS

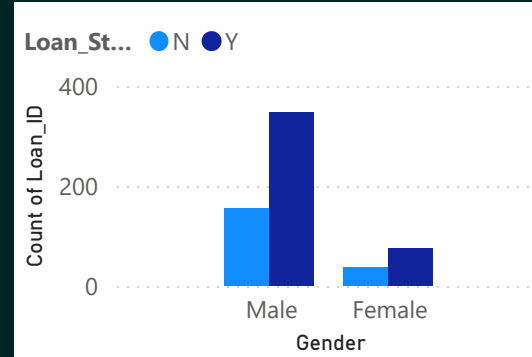
KEY ANALYSIS

OVERALL LOAN APPROVAL TRENDS

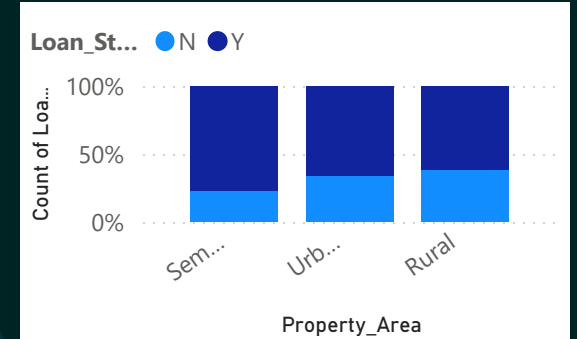
Count of Loan_ID by Loan_Status



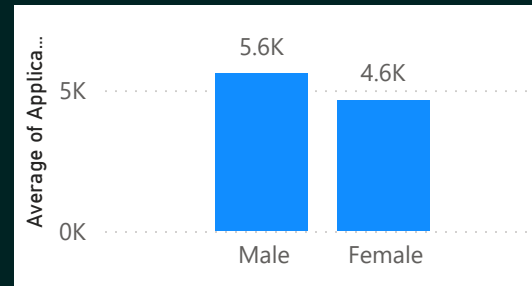
LOAN APPROVAL % BY GENDER



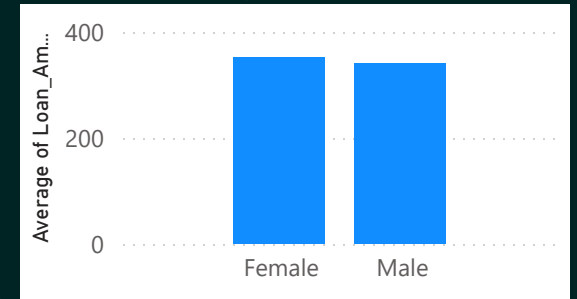
APPROVAL RATE BY PROPERTY AREA



AVERAGE APPLICANT INCOME BY GENDER



AVERAGE LOAN AMOUNT BY GENDER



NOTES

Applicants from Semiurban areas have the highest loan approval rate.

NOTES

Female applicants show a slightly higher loan approval percentage compared to male applicants.

BUSINESS PROBLEM:

WHICH GROUP SHOULD THE BANK PRIORITIZE FOR FASTER LOAN APPROVALS

INSIGHTS:

- Applicants with CREDIT_HISTORY = 1 had a 79,6% approval rate.
- Applicants in SEMIURBAN areas had the highest approval rate (-85%).
- FEMALE applicants showed a slightly higher approval percentage than male applicants

RECOMMENDATION:

The bank should prioritize semi-urban applicants with good credit history (CREDIT_HISTORY = 1) for faster and more efficient loan

CONCLUSION

- Credit history is the strongest indicator of faster loan approval.
- Semiurban applicants achieve the highest approval rates.
- Female applicants only slightly outperform their male counterparts. Reflects relatively equal opportunity within the loan approval process
- Data suggests that focusing on credit reliability and semiurban applicants can streamline loan approval processes for efficiency and equity.,
- This analysis equips loan officers with data-backed criteria to prioritize approvals more efficiently.